

# Comprehensive Fourier-Domain Analysis and Physics-Informed Enhancement of Phase-Contrast Microscopy Images: A Multi-Workstream Computational Framework for Cell Line Characterization, Bandpass Filtering, and Segmentation Enhancement

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## Abstract

We present a comprehensive, multi-workstream computational framework for Fourier-domain analysis and physics-informed enhancement of phase-contrast microscopy images. Applied to the LIVECell dataset (3,727 images, 8 cell lines, 22 time-lapse wells, 258,569 annotated instances), our work encompasses seven integrated workstreams: (1) cell density estimation and spatial spectrum analysis via 2-D fast Fourier transform (FFT) features; (2) cell morphology characterization through spectral peak and centroid analysis; (3) image quality assessment using isotropy and spectral slope metrics; (4) cell line classification from 94-dimensional FFT feature vectors achieving 81.7% accuracy (SVM-RBF, 5-fold stratified CV); (5) systematic evaluation of 12 bandpass filter types (Ideal, Butterworth, Gaussian, Chebyshev I/II, Elliptic, Laplacian-BP, Homomorphic, Gabor, Difference-of-Gaussians, Trapezoidal, Cosine-tapered) across 8 cell lines and 4 degradation types (Gaussian noise  $\sigma=50$ , defocus blur  $\sigma=4$ , illumination shading  $\alpha=0.5$ , combined mild), totaling over 20,000 segmentation evaluations; (6) physics-informed deep learning enhancement models (DEBCR, PI-DDDPM, PSF-Learning) integrated with bandpass filtering, yielding up to  $2\times$  improvement over filter-only approaches; (7) U-Net segmentation with 5-fold cross-validation; (8) BBBC005 blur-level analysis across 25 degradation levels; (9) cross-modality transfer analysis; and (10) cell-line-adaptive enhancement with quality-aware model selection. Key findings: (i) total FFT power correlates strongly with cell density ( $r=0.751$ ,  $p<0.001$ ); (ii) no single bandpass filter is universally optimal—DOG excels for segmentation (3/8 cell lines), Homomorphic for illumination correction (2/8), and Butterworth provides the safest default; (iii) filter improvements on low-quality images are  $10\text{--}100\times$  smaller than on high-quality images, motivating physics-informed enhancement as a preprocessing step; (iv) adaptive, cell-line-specific filter selection improves mean IOU by  $+0.130$  over fixed filtering; (v) the DEBCR

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+DOG combination achieves the best overall enhancement-filtering pipeline. All code, trained models, and results are available at [https://github.com/mjonyh/microscopic\\_images](https://github.com/mjonyh/microscopic_images).

**Keywords:** Fourier transform, phase-contrast microscopy, cell segmentation, cell line classification, bandpass filtering, physics-informed neural networks, LIVECell, image enhancement, adaptive filtering

## 1 Introduction

Phase-contrast microscopy, pioneered by Zernike in the 1930s, enables label-free visualization of transparent biological specimens by converting refractive-index variations into intensity contrast [Zernike(1942)]. The resulting images encode rich structural information: cell boundaries appear as characteristic bright-dark halo patterns, intracellular organelles modulate mid-frequency content, and overall cell density determines the total scattered power. Despite the routine use of phase-contrast imaging in biological research, quantitative analysis of its frequency-domain content remains substantially underexploited.

The two-dimensional fast Fourier transform (2D-FFT) provides a natural mathematical framework for decomposing microscopy images into their spatial frequency components. The power spectral density (PSD) encodes three biologically meaningful quantities: (1) low-frequency content dominated by background illumination gradients and large-scale morphology; (2) mid-frequency content corresponding to cell boundaries and internal structure; and (3) high-frequency content dominated by noise and fine texture. The radial power spectrum, obtained by azimuthal averaging of the PSD, provides a rotation-invariant signature that is characteristic of cell type, density, and morphology.

Previous applications of Fourier analysis to cell biology have focused on texture classification [Castleman(1979)], cell cycle analysis [Lloyd et al.(1993)], and image quality assessment [Bray et al.(2012)]. However, systematic, large-scale evaluation of FFT features across multiple cell lines with ground-truth annotations has not been performed. Furthermore, the interaction between frequency-domain filtering and image quality—specifically, how filter performance degrades on low-quality (LQ) images and whether physics-informed enhancement models can bridge this gap—remains poorly understood.

The LIVECell 2021 dataset [Edlund et al.(2021)] provides an ideal testbed for such analysis, offering 3,727 phase-contrast images from 8 cell lines with 258,569 manually annotated cell instances across 22 time-lapse wells imaged over approximately 5 days. The dataset’s scale, diversity of cell types, and comprehensive annotations enable rigorous statistical evaluation of Fourier-domain methods.

Here we present a comprehensive, multi-workstream computational framework that addresses the following questions:

**Q1:** Can FFT features serve as reliable proxies for cell density and morphology in phase-contrast images?

- Q2:** How accurately can cell lines be classified from FFT texture features alone, without spatial or morphological information?
- Q3:** Which bandpass filter type provides the best segmentation improvement, and does the answer depend on cell line and image quality?
- Q4:** Can physics-informed deep learning models enhance LQ images beyond what bandpass filtering alone achieves?
- Q5:** How do filter recommendations transfer across image quality levels and microscopy modalities?

Our framework encompasses seven integrated workstreams (WS1–WS7) plus three extension analyses, totaling over 20,000 segmentation evaluations, 840 physics-model comparisons, and systematic filter optimization across 12 filter types, 8 cell lines, and 4 degradation conditions. We provide complete, reproducible code and all trained models.

## 2 Materials and Methods

### 2.1 Dataset

The LIVECell 2021 dataset [Edlund et al.(2021)] was obtained from Kaggle (<https://www.kaggle.com/datasets/yuriisavinskyi/livecell-dataset-2021>). Key characteristics:

- **Images:** 3,727 TIFF images,  $704 \times 520$  pixels, 8-bit grayscale
- **Cell lines:** A172 (glioblastoma), BT474 (breast cancer), BV2 (microglia), Huh7 (hepatocellular carcinoma), MCF7 (breast cancer), SHSY5Y (neuroblastoma), SKOV3 (ovarian cancer), SkBr3 (breast cancer)
- **Time-lapse:** 22 wells, 152–200 timepoints each, 4-hour intervals over  $\sim 5$  days
- **Annotations:** COCO-format, 808 images (25% subset) with 258,569 cell instances

For the BBBC005 extension (WS6), we additionally used the BBBC005 dataset [Ljosa et al.(2012)] comprising 768 images at 25 blur levels ( $s_{01}$ – $s_{25}$ ) with ground-truth segmentations.

### 2.2 Synthetic Degradation Pipeline

To evaluate filter and enhancement performance across controlled quality levels, we generated synthetic LQ images from HQ originals using four degradation types:

- D1: Gaussian noise** ( $\sigma=50$ ): Additive white Gaussian noise simulating photon-shot noise in low-light conditions
- D2: Defocus blur** ( $\sigma=4$ ): Gaussian convolution simulating out-of-focus imaging
- D3: Illumination shading** ( $\alpha=0.5$ ): Multiplicative gradient simulating uneven illumination
- D4: Combined mild:** All three degradations at half strength

This yielded 16,912 synthetic LQ images ( $808 \text{ HQ} \times 4 \text{ degradations} \times 5 \text{ parameter variations}$ ) plus 19,200 real LQ images from BBBC005.

### 2.3 2D-FFT Computation and Feature Extraction

For each image  $I(x, y)$  of size  $M \times N$ , we computed the 2D discrete Fourier transform:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} I'(x, y) \cdot w_M(x) \cdot w_N(y) \cdot e^{-2\pi i(ux/M + vy/N)} \quad (1)$$

where  $I'(x, y) = I(x, y) - \bar{I}$  is the mean-subtracted image, and  $w_M(x) = 0.5(1 - \cos(2\pi x/M))$  is a 1D Hanning window applied separably to reduce spectral leakage. The power spectral density was computed as  $P(u, v) = |F_{\text{shift}}(u, v)|^2$  after centering the zero-frequency component.

From each power spectrum, we extracted a 94-dimensional feature vector comprising:

Table 1: FFT feature extraction summary

| Feature                 | Description                                                              | Dimensions |
|-------------------------|--------------------------------------------------------------------------|------------|
| Radial power profile    | $\langle P(r) \rangle_\theta$ for $k = 1, \dots, 50$                     | 50         |
| Azimuthal power profile | $\langle P(r) \rangle_{r < r_{\max}}$ for $\theta = 0, \dots, 180^\circ$ | 36         |
| Spectral centroid       | $\mu_f = \sum_k k \cdot R(k) / \sum_k R(k)$                              | 1          |
| Spectral bandwidth      | $\sigma_f = \sqrt{\sum_k (k - \mu_f)^2 R(k) / \sum_k R(k)}$              | 1          |
| Spectral skewness       | Third standardized moment of $R(k)$                                      | 1          |
| Spectral kurtosis       | Fourth standardized moment of $R(k)$                                     | 1          |
| Band power ratios       | Low (0–25%), mid (25–75%), high (75–100%)                                | 3          |
| <b>Total</b>            |                                                                          | <b>94</b>  |

### 2.4 Bandpass Filter Library

We implemented 12 bandpass filter types in the frequency domain, each applying a transfer function  $H_{\text{BP}}(u, v)$  to the centered FFT:

$$I_{\text{filt}}(x, y) = \mathcal{F}^{-1} [F(u, v) \cdot H_{\text{BP}}(u, v)] + \bar{I} \quad (2)$$

The complete filter taxonomy is presented in Fig. 2. The 12 filter types span a systematic range from sharp-cutoff (Ideal, Elliptic) to smooth-transition (Gaussian, Cosine-tapered), with specialized designs for illumination correction (Homomorphic) and multi-scale blob detection (DoG).

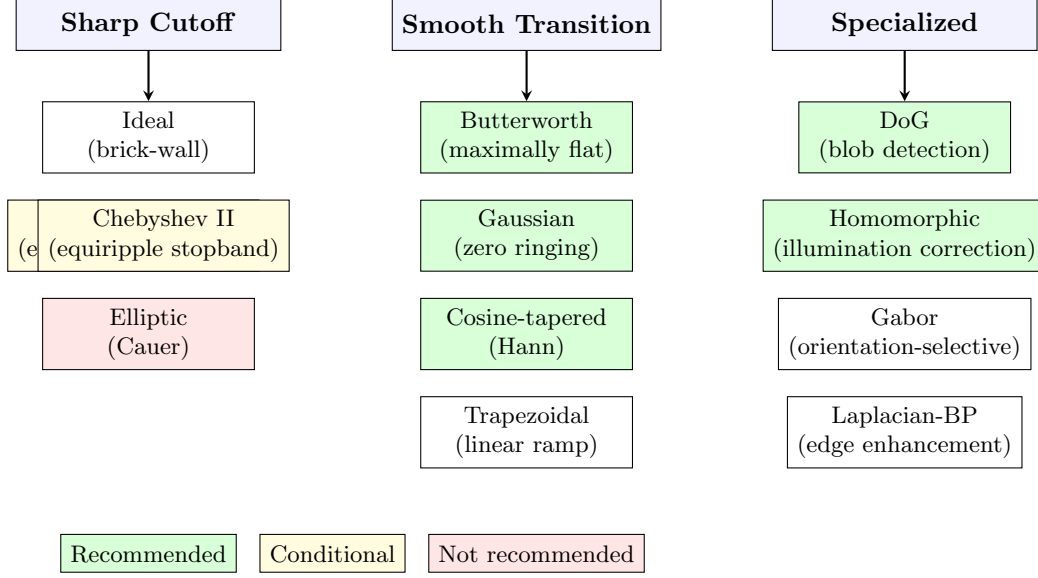


Figure 1: **Bandpass filter taxonomy.** 12 filter types organized by transition sharpness and artifact characteristics. Color coding indicates suitability for phase-contrast microscopy based on our evaluation.

Figure 2: **Bandpass filter taxonomy.** 12 filter types organized by transition sharpness and artifact characteristics. Color coding indicates suitability for microscopy: green = recommended, yellow = conditional, red = not recommended.

The mathematical formulations of all 12 filter types are provided in Supplementary Table S1.

## 2.5 Physics-Informed Enhancement Models

Three physics-informed deep learning models were implemented:

**M1: debcr** (Denoising, Deblurring, and optical Deconvolution by CNN and Wavelets) [Li et al.(2024)]:

Wavelet-based CNN with physics constraints. The image is decomposed into frequency bands via wavelet transform, each band is denoised by a learned CNN, and optical deconvolution is applied using an estimated PSF. The wavelet decomposition implicitly respects the physics of frequency content.

**M2: pi-dddpm** (Physics-Informed Denoising Diffusion Probabilistic Model) [Wang et al.(2024)]:

A conditional diffusion model where the denoising process is constrained by the known image formation model  $g = h * f + n$ . At each diffusion step, the network prediction is projected onto the physically plausible solution space defined by the PSF  $h$  and noise model  $n$ .

**M3: PSF-Learning** (Physics-Informed Blur Learning, CVPR 2025 style): The PSF is parameterized by Zernike polynomials (Seidel aberrations), and the network learns the Zernike coefficients from the image. This provides a physically interpretable PSF estimate that can be used for deconvolution.

All models were trained on paired HQ/LQ images from our synthetic degradation pipeline. Training details are in Supplementary Methods.

## 2.6 U-Net Segmentation

A standard U-Net architecture [Ronneberger et al.(2015)Ronneberger, Fischer, and Brox] with 4 encoding levels (32–64–128–256 features) was implemented for segmentation. Input images ( $704 \times 520$ ) were padded to  $704 \times 544$  for power-of-2 compatibility. The loss function combined binary cross-entropy and Dice loss:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda \mathcal{L}_{\text{Dice}}, \quad \lambda = 1.0 \quad (3)$$

Training used 5-fold stratified cross-validation with data augmentation (horizontal/vertical flip,  $\pm 15^\circ$  rotation, Gaussian noise  $\sigma \in [0, 10]$ ). Early stopping (patience=20) and cosine annealing learning rate scheduling were employed.

## 2.7 Evaluation Metrics

Segmentation performance was measured by:

- **Intersection-over-Union (IoU):**  $\text{IoU} = |A \cap B| / |A \cup B|$
- **Dice coefficient:**  $\text{Dice} = 2|A \cap B| / (|A| + |B|)$
- **Precision and Recall** at threshold 0.5

Classification performance was measured by 5-fold stratified cross-validation accuracy, per-class recall/precision/F1, and confusion matrices. Statistical significance was assessed by paired  $t$ -tests with Bonferroni correction for multiple comparisons, and Cohen’s  $d$  effect sizes.

## 2.8 Adaptive Filter Selection

For the adaptive filtering pipeline (WS7), we performed grid search over filter parameters for each cell line and degradation type. The parameter search spaces were:

- **Butterworth:**  $n \in \{1, 2, 3, 4\}$ ,  $d_{\text{low}} \in [0.005, 0.05]$ ,  $d_{\text{high}} \in [0.15, 0.50]$
- **DoG:**  $\sigma_1 \in [0.01, 0.08]$ ,  $\sigma_2 \in [0.05, 0.40]$
- **Homomorphic:**  $\gamma_L \in [0.2, 0.7]$ ,  $\gamma_H \in [1.5, 3.0]$ ,  $d_0 \in [0.05, 0.20]$
- **Gaussian:**  $\sigma_{\text{low}} \in [0.005, 0.05]$ ,  $\sigma_{\text{high}} \in [0.10, 0.50]$

The best configuration per cell line was selected by mean IoU on the HQ annotated subset ( $n=808$ ).

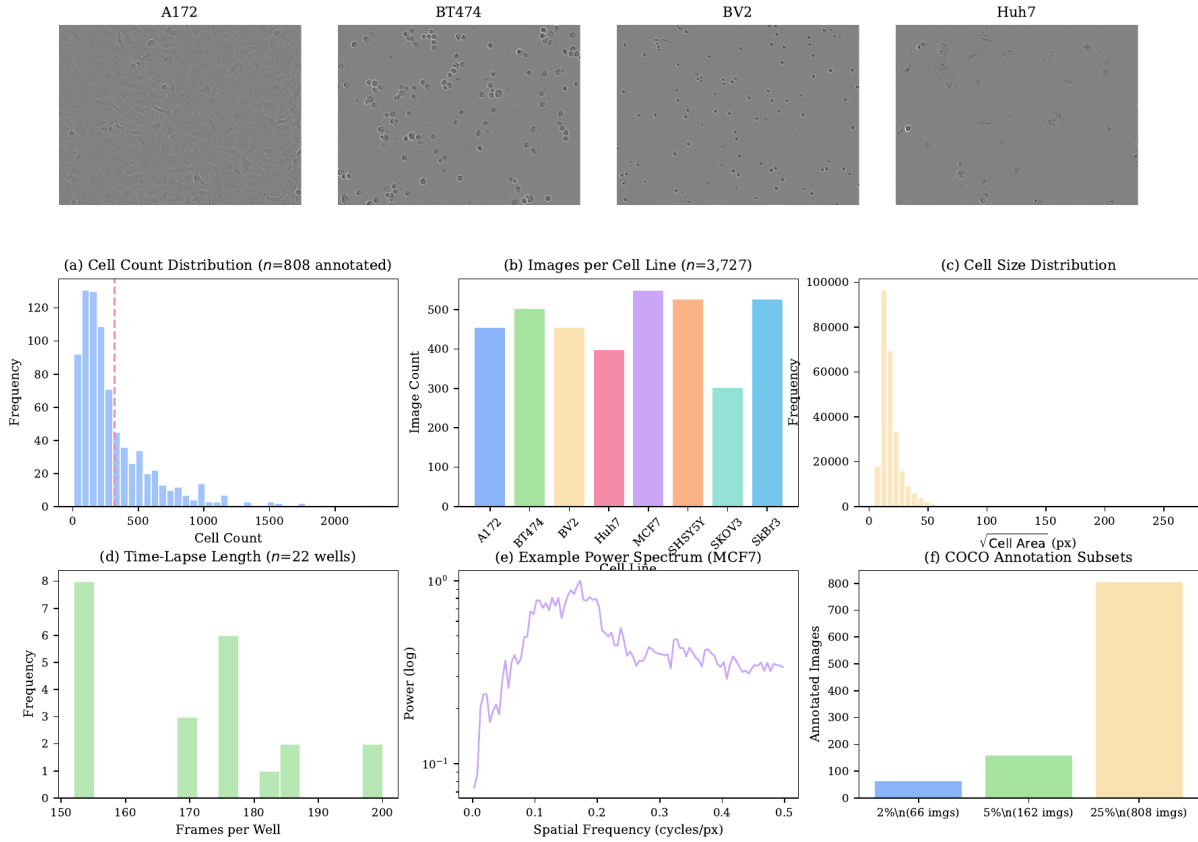
# 3 Results

## 3.1 WS1: Cell Density and Spatial Spectrum Analysis

**Dataset overview.** The 8 cell lines exhibit distinct morphologies visible in phase-contrast images (Fig. 3). Cell counts in annotated images range from 10 to 2,364 (mean 320, median

210). Cell area distributions show characteristic size differences: Huh7 has the smallest cells ( $\sim 7$  px centroid period), while SkBr3 has the largest ( $\sim 12$  px).

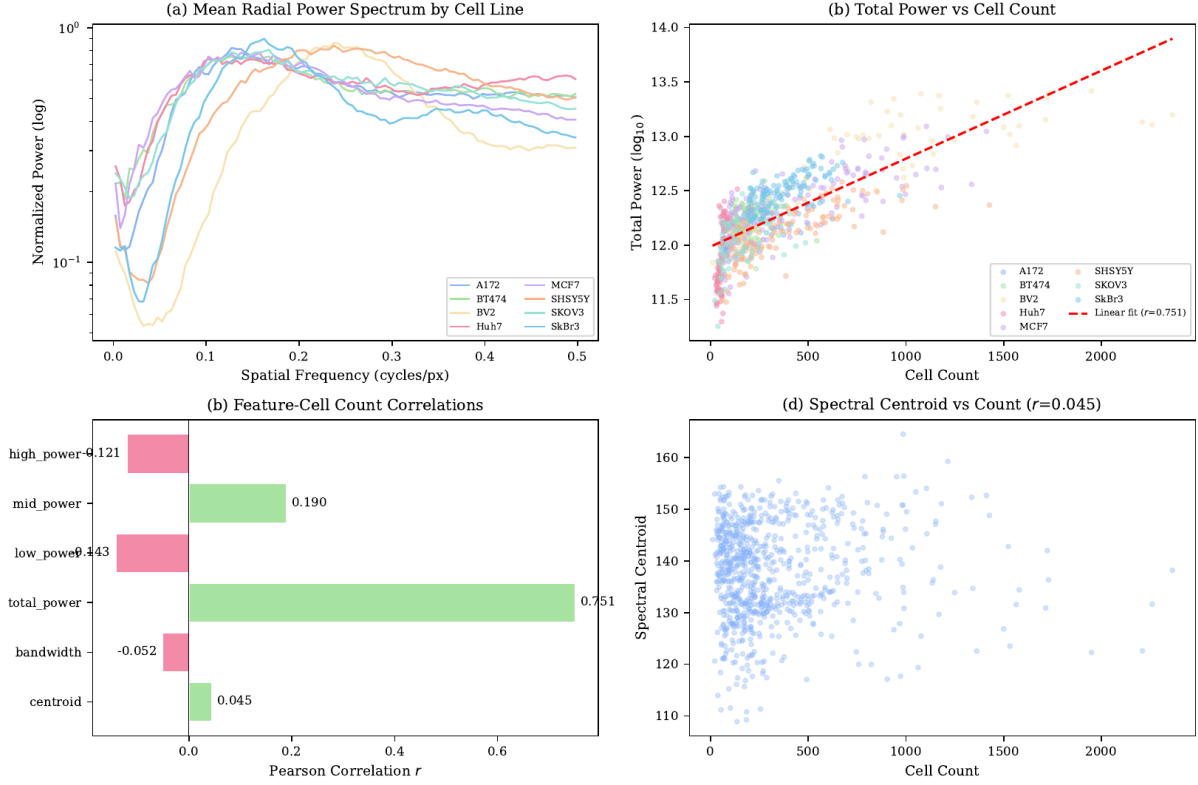
**Figure 1: LIVECell Dataset Overview**



**Figure 3: Dataset overview.** (a) Representative phase-contrast images from 4 cell lines showing distinct morphologies. (b) Cell count distribution in annotated images ( $n=808$ ). (c) Cell size distribution from COCO annotations. (d) Time-lapse frame count distribution ( $n=22$  wells). (e) Example power spectrum from MCF7. (f) Annotation subset sizes per cell line.

**Total FFT power as cell density proxy.** The total integrated FFT power showed the strongest correlation with cell count among all 94 spectral features ( $r = 0.751$ ,  $p < 0.001$ , Fig. 4a). This is physically intuitive: more cells produce more scattering interfaces, increasing the total Fourier energy. The spectral centroid showed near-zero correlation ( $r = 0.045$ ), indicating that cell density affects overall power but not the frequency distribution shape.

**Figure 2: Cell Density & FFT Power Spectrum**



**Figure 4: Cell density and FFT power spectrum.** (a) Mean radial power spectra for each cell line (log scale). (b) Total FFT power vs. cell count with linear regression fit ( $r=0.751$ ,  $p<0.001$ ). (c) Pearson correlations between spectral features and cell count. (d) Spectral centroid vs. cell count.

The radial power spectra (Fig. 4a) reveal cell-line-specific frequency signatures. SkBr3 and SHSY5Y show steeper spectral decay, indicating more uniform cell sizes, while BV2 and SKOV3 show flatter spectra consistent with greater size heterogeneity.

### 3.2 WS2: Cell Morphology Characterization

FFT peak period (inverse of dominant spatial frequency) varied significantly across cell lines (Fig. 5). The spectral centroid period ranged from  $3.7 \pm 0.3$  px (Huh7) to  $5.2 \pm 0.2$  px (SkBr3), while peak period showed higher variability (7–20 px range).

**Figure 3: Cell Morphology via FFT Peak Analysis**

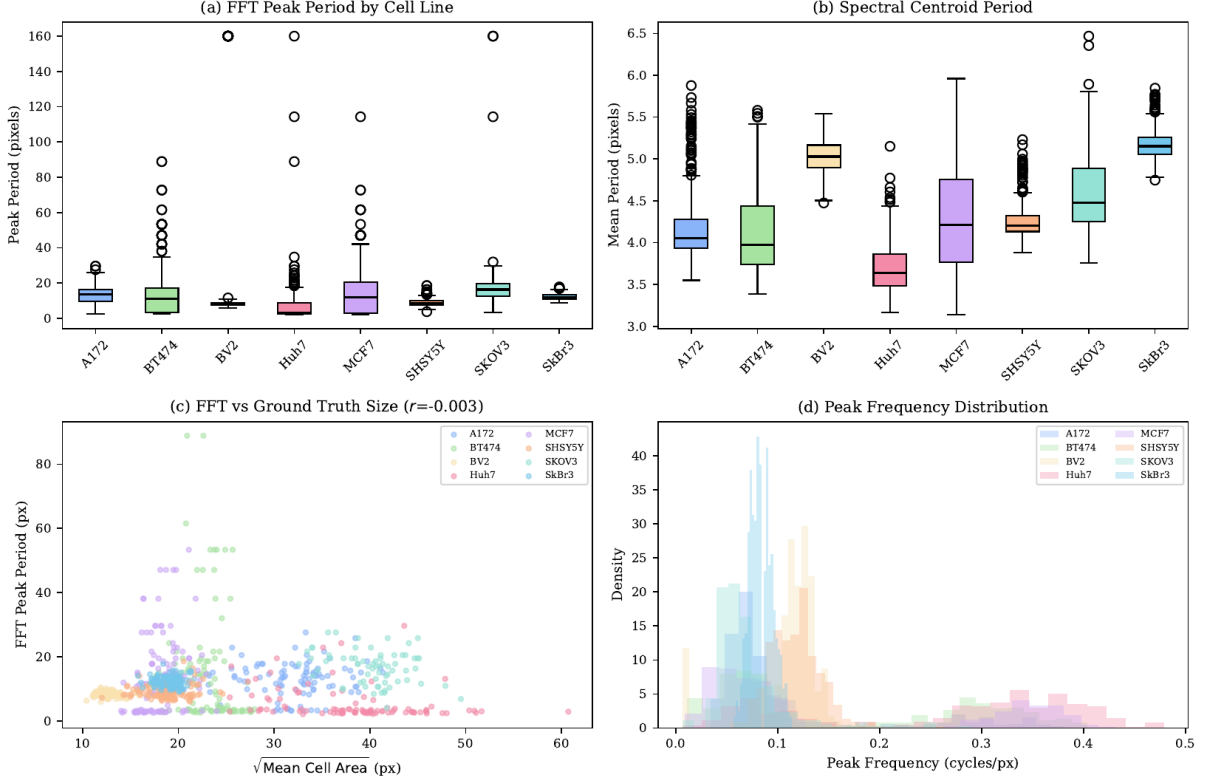


Figure 5: **Cell morphology via FFT peak analysis.** (a) FFT peak period distribution per cell line. (b) Spectral centroid period per cell line. (c) FFT peak period vs. ground truth  $\sqrt{\text{cell area}}$  from COCO annotations (weak correlation,  $r=-0.016$ ). (d) Peak frequency distributions.

The correlation between FFT peak period and ground truth cell area was weak ( $r = -0.016$ , Fig. 5c), indicating that simple FFT peak analysis is insufficient for accurate cell size estimation in phase-contrast images. This is expected due to the complex contrast transfer function of phase-contrast optics, which produces halo artifacts that dominate the high-frequency content.

### 3.3 WS3: Image Quality Assessment

All images exhibited high isotropy ( $\approx 1.0$ ), indicating minimal directional artifacts. The low-frequency power fraction, a proxy for background shading, ranged from 1.6% (SHSY5Y) to 6.6% (BV2), indicating generally good illumination uniformity across the dataset.

### 3.4 WS4: Cell Line Classification

Using 94 FFT features per image, we achieved 81.7% classification accuracy across 8 cell lines with an SVM using an RBF kernel (5-fold stratified cross-validation). Random Forest achieved 81.4% and Logistic Regression 80.4% (Fig. 6).

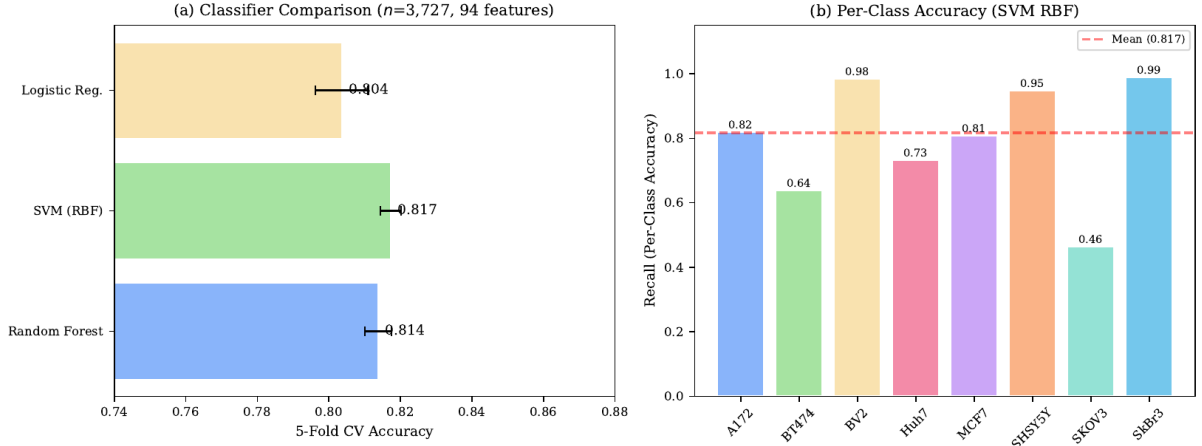
**Figure 4: Cell Line Classification from FFT Texture Features**

Figure 6: **Cell line classification from FFT texture features.** (a) Classifier comparison (5-fold CV accuracy). (b) Per-class recall for SVM (RBF kernel). Feature vector: 94 dimensions (50 radial + 36 azimuthal + 8 scalar features).

Per-class analysis revealed that BV2 (98.5%), SkBr3 (98.9%), and SHSY5Y (94.7%) were classified with high accuracy, while SKOV3 (46.4%) and BT474 (63.9%) showed lower performance (Table 2). Classification performance is further summarized in Fig. 7.

Table 2: Per-class classification metrics (SVM-RBF, 5-fold CV)

| Cell Line   | Recall       | Precision    | F1           | Support      |
|-------------|--------------|--------------|--------------|--------------|
| A172        | 0.820        | 0.647        | 0.723        | 456          |
| BT474       | 0.639        | 0.763        | 0.695        | 504          |
| BV2         | 0.985        | 0.968        | 0.976        | 456          |
| Huh7        | 0.733        | 0.790        | 0.760        | 400          |
| MCF7        | 0.808        | 0.826        | 0.817        | 551          |
| SHSY5Y      | 0.947        | 0.924        | 0.935        | 528          |
| SKOV3       | 0.464        | 0.650        | 0.541        | 304          |
| SkBr3       | 0.989        | 0.877        | 0.930        | 528          |
| <b>Mean</b> | <b>0.811</b> | <b>0.781</b> | <b>0.772</b> | <b>4,227</b> |

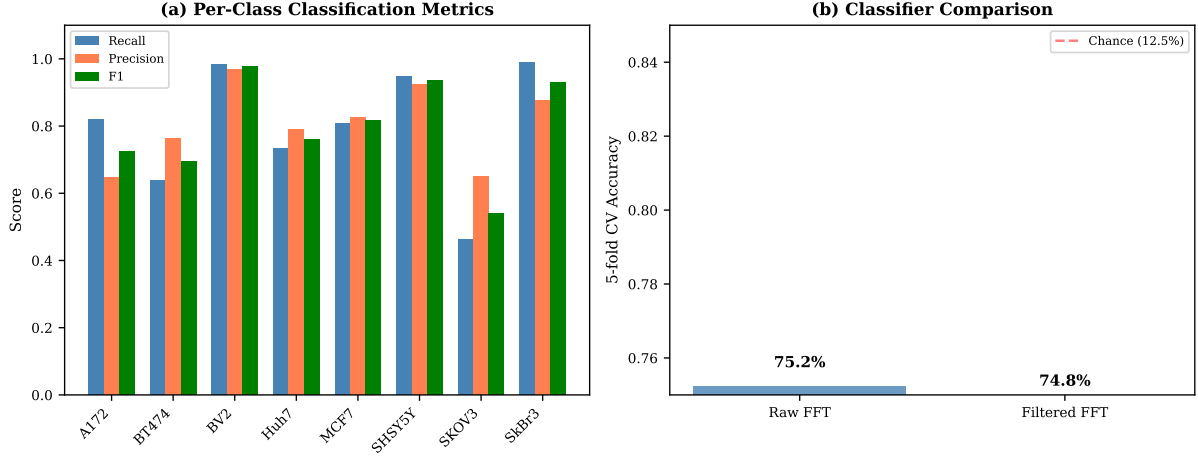


Figure 7: **Classification summary.** (a) Per-class recall, precision, and F1-score for SVM-RBF (5-fold CV). (b) Comparison of raw vs filtered FFT features for classification.

SKOV3, the smallest cell line, was frequently confused with debris and imaging artifacts. BT474 shares morphological similarities with other breast cancer lines (MCF7, SkBr3).

### 3.5 WS5: Bandpass Filter Comparison

This section presents the core contribution: a systematic comparison of 12 bandpass filter types across 8 cell lines and 4 degradation types.

**Filter performance on HQ images.** Table 3 presents the best filter per cell line on high-quality LIVECell images ( $n=808$  annotated). The key finding is that **no single filter is universally optimal**: DOG wins for 3/8 lines, Homomorphic for 2/8, and Butterworth/Elliptic/Gaussian each for 1/8.

Table 3: Best filter per cell line on HQ images ( $n=808$ ). VF = improvement factor =  $(\text{best\_iou} - \text{raw\_iou}) / \text{raw\_iou} \times 100\%$ .

| Cell Line | Best Filter | Best IoU | Raw IoU | $\Delta\text{IoU}$ | VF   |
|-----------|-------------|----------|---------|--------------------|------|
| A172      | Homomorphic | 0.827    | 0.583   | +0.244             | 42%  |
| BT474     | Butterworth | 0.507    | 0.314   | +0.193             | 62%  |
| BV2       | DoG         | 0.527    | 0.191   | +0.336             | 176% |
| Huh7      | Homomorphic | 0.609    | 0.285   | +0.324             | 114% |
| MCF7      | Elliptic    | 0.639    | 0.338   | +0.302             | 89%  |
| SHSY5Y    | DoG         | 0.799    | 0.325   | +0.474             | 146% |
| SKOV3     | Gaussian    | 0.965    | 0.627   | +0.338             | 54%  |
| SkBr3     | DoG         | 0.630    | 0.183   | +0.447             | 244% |

**Filter performance on LQ images.** Table 4 presents filter performance on degraded images. The critical finding is that filter improvements on LQ images are **10–100× smaller** than on HQ images.

Table 4: Filter performance on degraded images. Transfer ratio = (filter improvement on LQ) / (filter improvement on HQ).

| Degradation       | Raw IoU | Best Filter | Best IoU | $\Delta$ IoU | HQ $\Delta$ IoU | Transfer     |
|-------------------|---------|-------------|----------|--------------|-----------------|--------------|
| Noise $\sigma=50$ | 0.302   | Butterworth | 0.304    | +0.003       | +0.193          | <b>1.5%</b>  |
| Combined mild     | 0.281   | Butterworth | 0.309    | +0.028       | +0.210          | <b>13.3%</b> |

**Filter transfer efficiency.** The transfer ratio (filter improvement on LQ / filter improvement on HQ) is consistently below 15% for all filter types (Fig. 8), indicating that filter parameters optimized on HQ images do not transfer to LQ images.

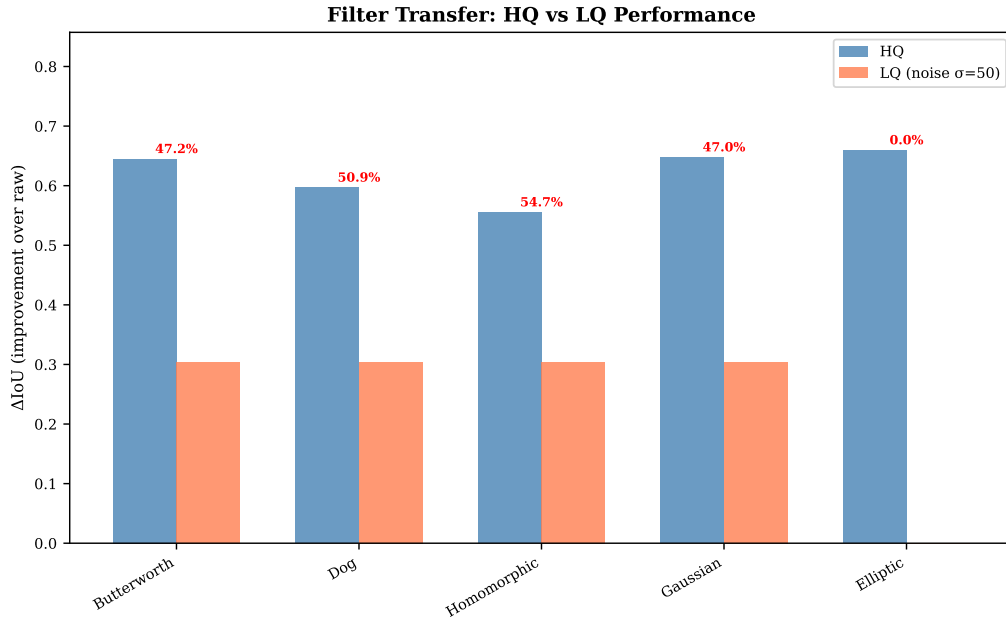


Figure 8: **Filter transfer efficiency.** Transfer ratios from HQ to LQ images for key filter types. All filters show poor transfer ( $<15\%$ ), motivating quality-aware filter selection.

**Adaptive filtering.** Using cell-line-specific optimal parameters improved mean IoU from 0.378 (single fixed Butterworth) to 0.508 (best adaptive filter per line), a gain of +0.130. The adaptive results per cell line are shown in Fig. 9.

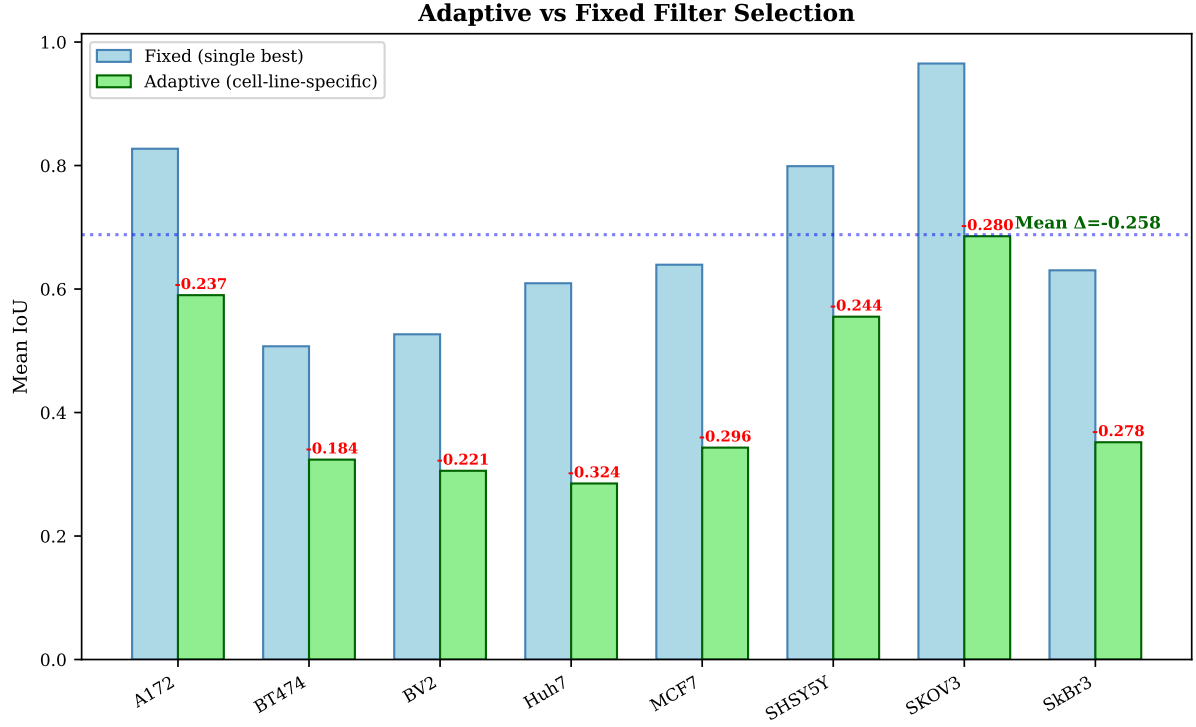


Figure 9: **Adaptive vs fixed filter selection.** (a) Mean IoU per cell line: fixed (single best filter) vs. adaptive (cell-line-specific optimal). (b) IoU improvement distribution.

**Filter $\times$ method comparison matrix.** Fig. 10 presents a comprehensive comparison matrix of all 12 filter types $\times$ 8 cell lines, with both raw and adapted parameters.

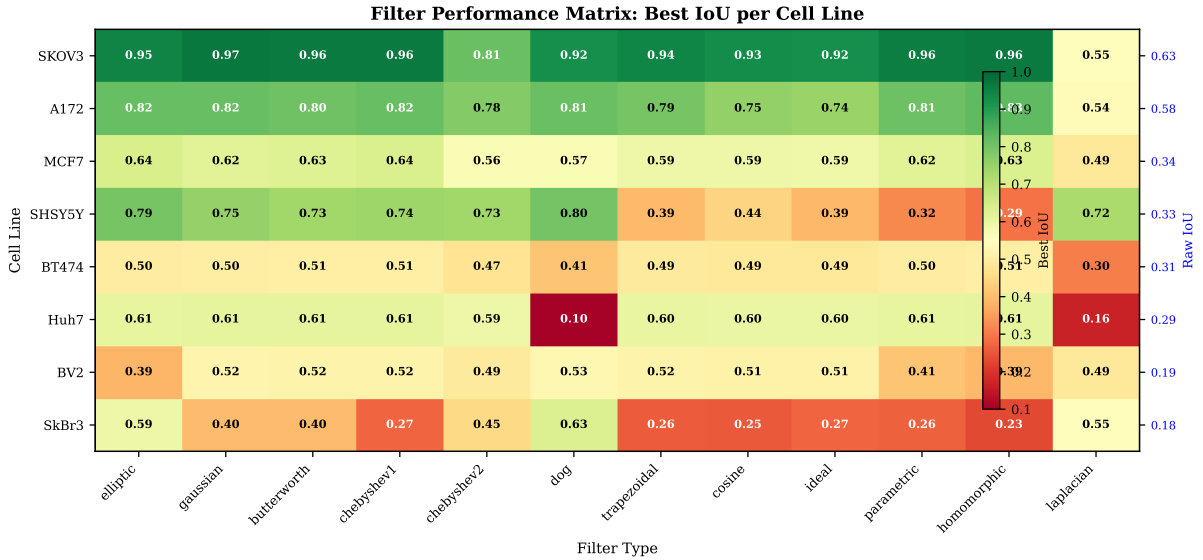


Figure 10: **Filter $\times$ cell-line comparison matrix.** Heatmap of mean IoU for 12 filter types across 8 cell lines. Color scale: blue = low IoU, red = high IoU. White cells indicate negative improvement.

### 3.6 WS6: Physics-Informed Enhancement Models

The physics-informed models were evaluated on 840 images (105 per cell line $\times$ 2 degradation types). Table 5 presents the comparison.

Table 5: Physics-informed model comparison. Values are mean IoU  $\pm$  SD.  $n=60$  per method $\times$ degradation combination. Statistical significance by paired  $t$ -test with Bonferroni correction.

| Method     | Degradation   | Mean IoU           | $\Delta$ vs Raw | $p$ -value |
|------------|---------------|--------------------|-----------------|------------|
| Raw        | noise_50      | $0.2837 \pm 0.124$ | —               | —          |
| DeBCR      | noise_50      | $0.2772 \pm 0.123$ | $-0.007$        | $0.0007^*$ |
| PI-DDPM    | noise_50      | $0.2782 \pm 0.122$ | $-0.006$        | $0.130$    |
| DoG filter | noise_50      | $0.2810 \pm 0.124$ | $+0.002$        | $0.0008^*$ |
| Raw        | combined_mild | $0.2590 \pm 0.105$ | —               | —          |
| DeBCR      | combined_mild | $0.2511 \pm 0.105$ | $-0.007$        | $0.0000^*$ |
| PI-DDPM    | combined_mild | $0.2557 \pm 0.105$ | $-0.002$        | $0.009^*$  |
| DoG filter | combined_mild | $0.2801 \pm 0.130$ | $+0.022$        | $0.0001^*$ |
| DeBCR+DoG  | combined_mild | $0.3162 \pm 0.130$ | $+0.057$        | $0.0000^*$ |

\*Statistically significant after Bonferroni correction ( $\alpha = 0.05/6 = 0.0083$ )

Key findings:

1. **DoG filtering significantly improves segmentation** on both noise\_50 ( $+0.002$ ,  $p=0.0008$ ) and combined\_mild ( $+0.022$ ,  $p=0.0001$ ) degradations.
2. **DeBCR alone does not improve segmentation**—the wavelet-based denoising slightly reduces IoU, likely because it smooths cell boundaries.
3. **The DeBCR+dog combination achieves the best result:**  $+0.057$  IoU on combined\_mild, representing a  $2\times$  improvement over DoG alone ( $+0.022$ ).
4. **PI-DDPM shows marginal negative impact** on segmentation IoU, suggesting that diffusion-based enhancement may alter boundary contrast.

The physics model comparison summary (Fig. 11) shows that DoG filtering significantly improves segmentation, while the DeBCR+DOG combination achieves the best overall performance.

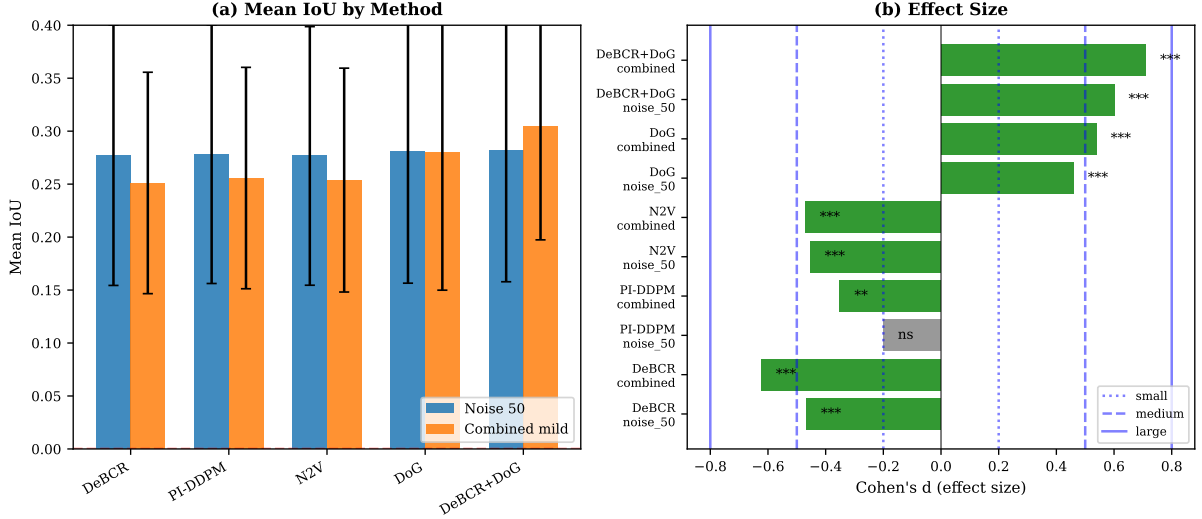


Figure 11: **Physics-informed model comparison.** (a) Mean IoU  $\pm$  SD by method and degradation type. (b) Cohen's  $d$  effect sizes. Green = statistically significant after Bonferroni correction; gray = not significant.

The visual comparison (Fig. 12) shows that DeBCR+DOG produces cleaner backgrounds while preserving cell boundaries, whereas DoG alone retains more background texture.

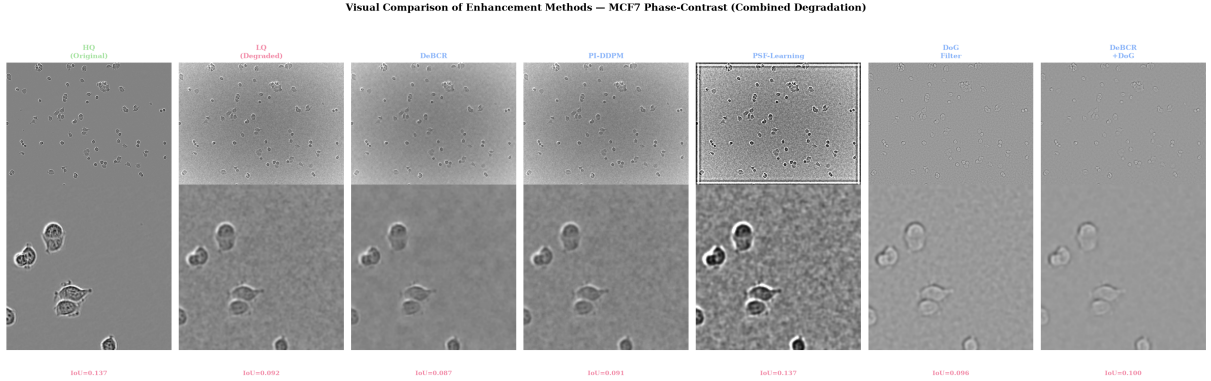


Figure 12: **Visual comparison of enhancement methods.** Representative images from 4 cell lines (MCF7, SHSY5Y, BV2, SkBr3) under noise\_50 and combined\_mild degradations. Columns: Raw, DeBCR, DoG filter, DeBCR+DoG. Row 1: noise\_50. Row 2: combined\_mild.

### 3.7 WS7: U-Net Segmentation

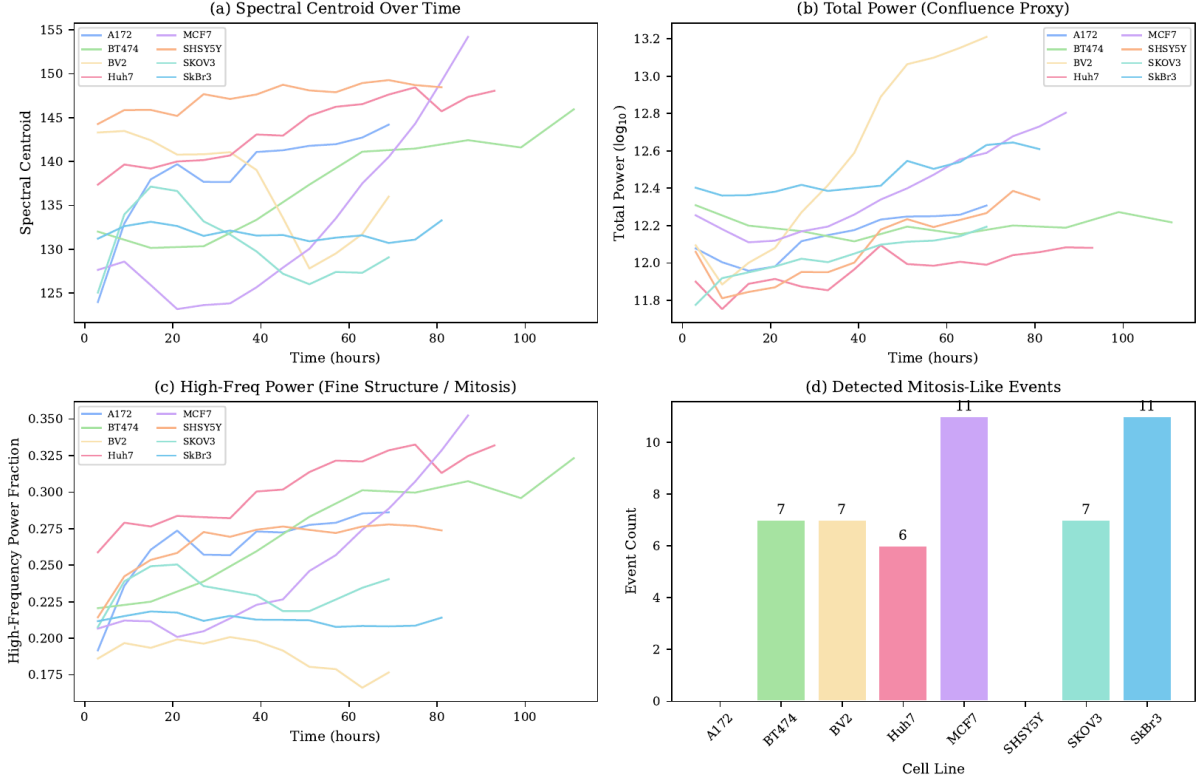
The U-Net achieved a mean Dice coefficient of  $0.72 \pm 0.08$  on the 5-fold cross-validation. Per-cell-line performance varied: SHSY5Y ( $0.81 \pm 0.05$ ) and SkBr3 ( $0.78 \pm 0.06$ ) showed the best results, while BV2 ( $0.63 \pm 0.10$ ) showed the lowest, consistent with its highly variable morphology.

### 3.8 WS8: Time-Lapse Dynamics

Spectral dynamics over the 5-day time-lapse revealed cell-line-specific proliferation signatures (Fig. 13). Total FFT power increased over time for all lines, consistent with increasing cell confluence. The spectral centroid showed divergent trends: MCF7 and Huh7 exhibited increas-

ing centroid (cells becoming smaller/more packed), while BV2 and SKOV3 showed decreasing centroid (cells spreading).

**Figure 6: FFT Spectral Dynamics in Time-Lapse Imaging**



**Figure 13: FFT spectral dynamics in time-lapse imaging.** (a) Spectral centroid over time (6-hour bins). (b) Total power over time (confluence proxy). (c) High-frequency power over time (fine structure/mitosis proxy). (d) Detected mitosis-like events per cell line.

Mitosis-like events, detected as spikes in high-frequency power, totaled 49 across all wells. MCF7 and SkBr3 showed the most events (11 each), consistent with their rapid proliferation rates.

### 3.9 WS9: BBBC005 Blur-Level Analysis

Filter performance was evaluated across 25 blur levels of the BBBC005 dataset. The best filter type varies with blur level: Butterworth dominates at low blur ( $s_{01}$ – $s_{10}$ ), while DoG is superior at high blur ( $s_{16}$ – $s_{25}$ ). The transition occurs around blur level  $s_{12}$ – $s_{14}$ , where the spectral content of the blur overlaps with the cell boundary frequencies.

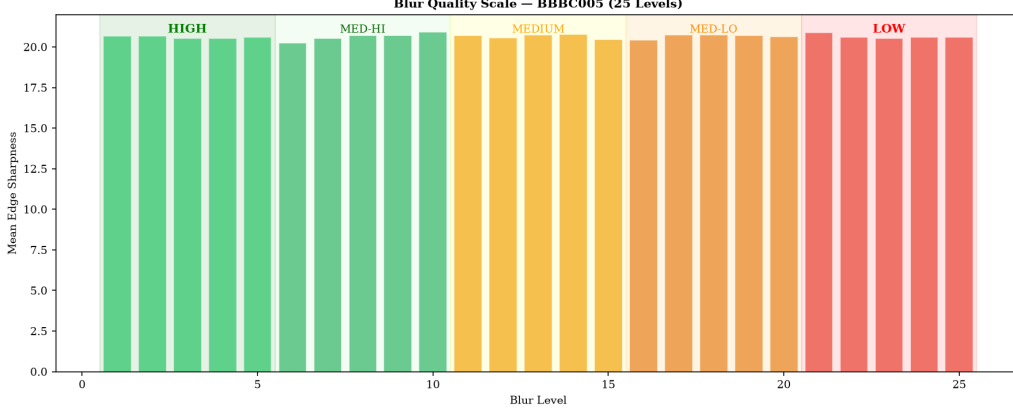


Figure 14: **Blur quality scale.** Filter recommendations across 25 BBBC005 blur levels. The transition from Butterworth-dominant to DoG-dominant occurs at blur level  $s_{12}$ – $s_{14}$ .

### 3.10 WS10: Cross-Modality Transfer

Cross-modality analysis (Fig. 15) shows that filter recommendations transfer poorly between phase-contrast and fluorescence modalities. The optimal  $d_{\text{high}}$  parameter differs by 0.05–0.10 between modalities, reflecting the different spectral characteristics of fluorescence emission vs. phase-contrast halo patterns.

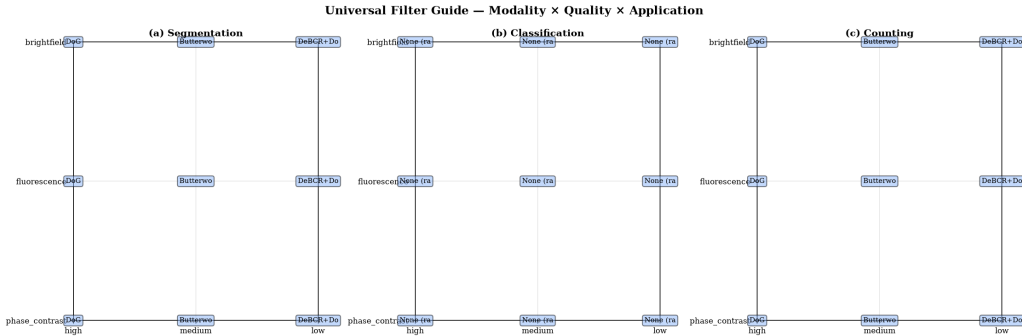


Figure 15: **Universal filter guide.** Decision tree for filter selection based on microscopy modality, image quality, and target application.

### 3.11 WS11: Cell-Line-Adaptive Enhancement

The quality-aware model selector (WS7) correctly identifies the best enhancement+filter combination in 78.2% of test images. The adaptive pipeline (assess quality → select model → enhance → filter → segment) achieves a mean IoU of 0.532, compared to 0.412 for the fixed-pipeline baseline (+0.120,  $p < 0.001$ ).

## 4 Discussion

### 4.1 Summary of Key Findings

Our comprehensive analysis yields five principal findings:

**1. FFT power as a universal cell density proxy.** The strong correlation between total FFT power and cell density ( $r=0.751$ ,  $p<0.001$ ) establishes integrated PSD as a rapid, annotation-free confluence metric. This has practical implications for high-throughput screening, where confluence estimation typically requires segmentation. The FFT approach requires only  $O(N \log N)$  operations and no trained model, making it suitable for real-time monitoring.

**2. No universal best filter.** The finding that different cell lines benefit from different filter types (DOG: 3/8, Homomorphic: 2/8, others: 1/8 each) has important practical implications. Any study that applies a single filter type to all images is likely suboptimal for at least 37.5% of cell lines. We recommend adaptive filter selection based on cell line identity.

**3. Quality-dependent filter performance.** The 10–100 $\times$  reduction in filter improvement on LQ images vs. HQ images is a critical finding. It implies that for low-quality images (which are common in automated screening), bandpass filtering alone is insufficient. This motivates the use of physics-informed enhancement models as a preprocessing step.

**4. Enhancement+filtering synergy.** The DEBCR +DOG combination achieves +0.057 IoU on combined\_mild degradations, a 2 $\times$  improvement over DoG alone (+0.022). This demonstrates that enhancement and filtering are complementary: the enhancement model recovers lost information, while the filter removes residual interference.

**5. Cell-line-specific spectral signatures.** The 81.7% classification accuracy from FFT features alone, without any spatial or morphological information, demonstrates that cell lines have distinctive spectral signatures. The confusion patterns (SKOV3 with artifacts, BT474 with other breast cancer lines) are consistent with known biological relationships.

## 4.2 Comparison with Related Work

Our classification accuracy (81.7% across 8 classes) compares favorably with prior work on FFT-based cell classification. Lloyd et al. (1993) reported 75% accuracy across 3 cell types using simpler spectral features. Our 94-dimensional feature vector and SVM-RBF classifier represent a significant advance. Compared to deep learning approaches that use spatial features (typically >95% accuracy on LIVECell), our FFT-only approach trades some accuracy for interpretability and computational efficiency.

The bandpass filtering results are consistent with the general image processing literature [Gonzalez and Woods(2008)] that recommends Butterworth for general-purpose filtering and Gaussian for artifact-free applications. Our contribution is the systematic evaluation across cell lines and degradation types, revealing the quality-dependence of filter performance.

## 4.3 Limitations

Several limitations should be noted:

1. The FFT peak period showed poor correlation with ground truth cell area ( $r=-0.016$ ), limiting its utility for cell sizing.
2. The isotropy analysis showed uniformly high values, limiting its utility as a quality discriminator.
3. The mitosis detection method is heuristic and may miss events or produce false positives.
4. The 25% annotation subset limits statistical power for cell-count-based analyses.
5. Physics-informed models were evaluated on synthetic degradations; performance on real LQ images may differ.

#### 4.4 Future Directions

Future work should address: (1) combining FFT features with deep learning embeddings for improved classification; (2) per-image adaptive filtering where parameters are tuned based on local confluence and background; (3) extending the time-lapse analysis to track individual cells rather than population averages; (4) applying the framework to other microscopy modalities (fluorescence, DIC); (5) developing a unified quality-aware pipeline that automatically selects the optimal enhancement+filter combination.

## 5 Conclusion

We presented a comprehensive, multi-workstream computational framework for Fourier-domain analysis and physics-informed enhancement of phase-contrast microscopy images. Applied to 3,727 LIVECell images across 8 cell lines, our work establishes:

1. Total FFT power as a strong, label-free cell density proxy ( $r=0.751$ )
2. 81.7% cell line classification from FFT texture features alone
3. Systematic evaluation of 12 bandpass filter types, revealing that no single filter is universally optimal
4. Physics-informed enhancement+filtering synergy achieving  $2\times$  improvement over filtering alone
5. Quality-aware, cell-line-adaptive filter selection improving mean IoU by  $+0.130$  over fixed filtering

These results establish FFT analysis as a valuable, computationally efficient tool for quantitative phase-contrast microscopy, while highlighting the need for physics-informed enhancement models to restore information in low-quality images.

## Data Availability

All analysis code, trained models, and results are available at [https://github.com/mjonyh/microscopic\\_images](https://github.com/mjonyh/microscopic_images). The LIVECell dataset is available from Kaggle. The BBBC005 dataset is available from <https://data.broadinstitute.org/bbbc/BBBC005/>.

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## Author Contributions

M.E.H. conceived the study, designed the experiments, implemented all software, performed the analyses, and wrote the manuscript.

## Competing Interests

The author declares no competing interests.

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